Vehicle Damage Classification Using Deep Learning

Final Project Report

Course: Deep Learning | Academic Project

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Abstract

This project evaluates convolutional neural networks for seven-category vehicle damage classification using photographs of damaged vehicles. After excluding unknown labels and deduplicating image references, 963 labeled images were used, with 674 training, 144 validation, and 145 held-out test images. A compact convolutional neural network (CNN) was compared with pretrained ResNet50 feature extractors trained without and with image augmentation. Recorded test accuracies were 42.07% for the baseline CNN, 77.93% for ResNet50 without augmentation, and 78.62% for augmented ResNet50. The augmented model achieved macro precision 0.81, macro recall 0.78, and macro F1-score 0.78. Its confusion matrix shows strong glass-shatter recognition but substantial confusion between door dents and door scratches. The study demonstrates the usefulness of transfer learning on a modest dataset while identifying limits in fine-grained damage discrimination and experimental comparability.

Keywords: vehicle damage; image classification; transfer learning; ResNet50; CNN; data augmentation; error analysis

1. Introduction

Vehicle damage assessment often requires distinguishing dents, scratches, and damage to glass or lighting components from photographs. Automated image classification could help triage inspection images and support downstream human review. The task is challenging because photographs vary in viewpoint, reflections, background, lighting, and the scale of visible damage. A scratch and a shallow dent can appear similar, particularly when the relevant region occupies only a small part of an image.

1.1 Problem and motivation

The research question is whether a pretrained image-classification backbone provides a useful performance improvement over a small CNN trained from scratch on the available labeled images, and whether adding geometric augmentation yields a further improvement in the recorded experiment. The focus is category classification rather than localization, repair-cost prediction, or severity estimation.

1.2 Objectives

- Prepare reproducible train, validation, and held-out test partitions for seven labeled classes.
- Train a lightweight baseline CNN and two ResNet50 variants.
- Evaluate held-out accuracy, per-class precision, recall, F1-score, and a confusion matrix.
- Inspect representative errors and discuss limitations and possible improvements.

1.3 Scope

The seven classes are bumper_dent, bumper_scratch, door_dent, door_scratch, glass_shatter, head_lamp, and tail_lamp. The labels identify damage categories, not necessarily damage extent or exact image coordinates. Results reported here are from the supplied notebook outputs and exported experiment artifacts; unobserved experiments are not represented as completed.

2. Dataset and Preprocessing

The uploaded source archive contains 1,512 image files and a CSV with 1,594 rows, including 549 entries labeled unknown. The experiment used 963 labeled images after excluding unknown entries and resolving repeated image references. The precise original public dataset URL was not included in the supplied files and should be added to the final bibliography when confirmed.

2.1 Class distribution

Category	Labeled CSV rows (before deduplication)
bumper_dent	129
bumper_scratch	164
door_dent	192
door_scratch	154
glass_shatter	137
head_lamp	133
tail_lamp	136

Note: row counts above describe source CSV records, not necessarily unique images. Repeated image references exist; train/validation/test sizes refer to the deduplicated labeled dataset used in the experiment.

2.2 Split strategy

Partition	Images	Approx. share
Training	674	70.0%
Validation	144	15.0%
Test	145	15.1%

The notebook workflow used stratified splitting to preserve the seven classes across partitions. The test set was held aside for final evaluation. All models were assessed against the recorded 145-image test partition.

2.3 Input handling

Images were resized to 224 × 224 pixels and grouped into TensorFlow batches of 16. The baseline CNN used a 1/255 rescaling layer. The augmented ResNet50 applied the model-specific ResNet50 preprocessing function after augmentation. These preprocessing paths are intentionally different because the baseline and pretrained backbone expect different input representations.

3. Model Architectures

3.1 Baseline CNN

The baseline architecture uses three convolutional blocks with 32, 64, and 128 filters, each followed by max pooling. Global average pooling reduces the final spatial representation; a 128-unit dense layer with ReLU and dropout (0.3) precedes the seven-class softmax. The model has 110,663 trainable parameters and was optimized with Adam using sparse categorical cross-entropy.

3.2 ResNet50 without augmentation

The non-augmented experiment used an ImageNet-pretrained ResNet50 backbone as a feature extractor for the seven-category task. The exported materials contain its recorded test accuracy (77.93%), model weights, and training history, but not the complete model configuration. This limits independent reconstruction of that run from the report materials alone.

3.3 ResNet50 with augmentation

The augmented architecture used ImageNet-pretrained ResNet50 with its backbone frozen, followed by global average pooling, a 128-unit ReLU dense layer, dropout (0.3), and a seven-way softmax. Training-time augmentation included random horizontal flips, rotations up to a factor of 0.08, and zoom changes up to a factor of 0.10. The augmentation layer is inactive at inference time. The optimizer was Adam with learning rate 0.001.

3.4 Experimental caution

The reported no-augmentation and augmentation results were collected across notebook runs that may not have identical preprocessing or training controls. The 0.69 percentage-point gap is a descriptive difference between recorded scores, not conclusive evidence of a causal augmentation benefit. A controlled comparison should use identical splits, seeds, optimizer settings, preprocessing, and backbone treatment, changing only augmentation.

4. Training Procedure and Model Selection

The baseline CNN and augmented ResNet50 were each trained for 15 epochs using the 674-image training partition and monitored against the 144-image validation partition. ModelCheckpoint saved the best model according to validation accuracy, and CSVLogger saved per-epoch loss and accuracy. The best checkpoints—not necessarily the last epoch—were used for testing.

Run	Epochs	Best validation accuracy
Baseline CNN	15	43.06%
ResNet50 + augmentation	15	80.56%

4.1 Interpretation of learning curves

The baseline CNN reached a best validation accuracy of 43.06%, while the augmented ResNet50 reached 80.56%. These maxima are descriptive summaries from the provided training-history CSVs. They are not interchangeable with held-out test accuracy, because validation data was used for checkpoint selection.

4.2 Reproducibility notes

The experiment outputs were saved to Google Drive as Keras checkpoints, CSV histories, and PNG figures. To reproduce the work independently, the notebook should specify the data filtering/deduplication rules, random seeds, exact preprocessing pipeline, train/validation/test identifiers, package versions, and the model-definition code for every experiment. Large datasets and model weights should be distributed separately from source code when repository size restrictions apply.

5. Held-Out Test Results

Model	Test accuracy	Test loss
Baseline CNN	42.07%	1.5848
ResNet50, no augmentation	77.93%	0.6126 (recorded earlier)
ResNet50, augmentation	78.62%	0.5573

The augmented ResNet50 attained 114 correct predictions out of 145 test examples, whereas the baseline CNN achieved approximately 61 correct predictions. Transfer learning yielded a substantial improvement over the small scratch-trained baseline in these runs. The difference between ResNet50 variants is only one test image (0.69 percentage points) and should not be overstated.

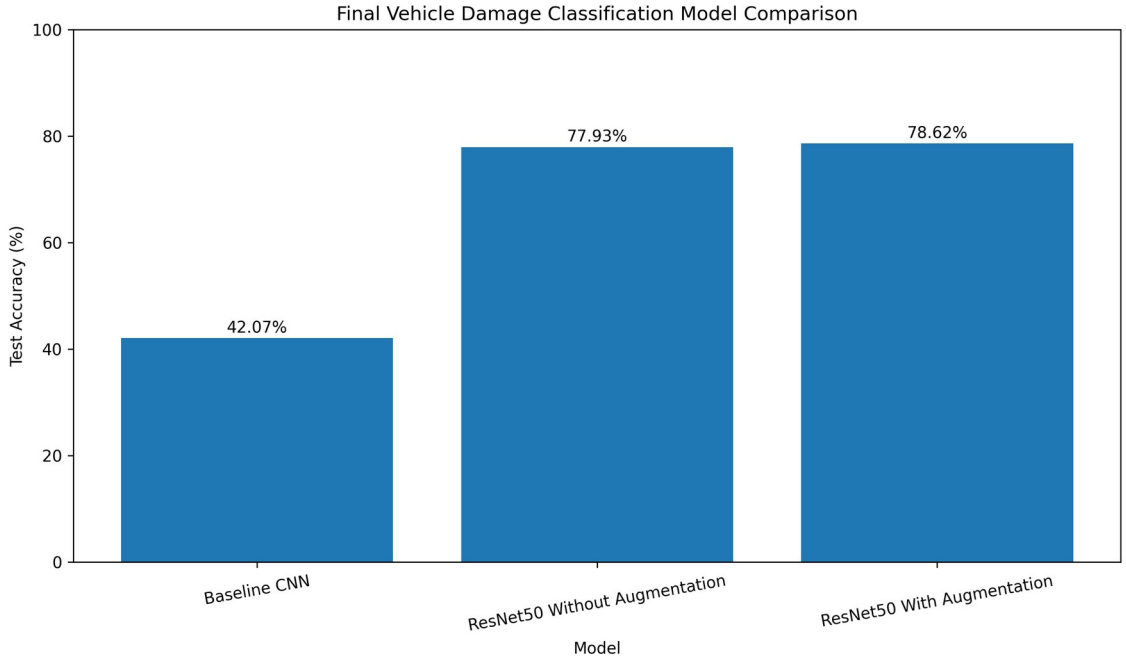


Figure 1. Held-out test accuracy comparison across the three trained models. The chart and table were exported from the user's experiment.

6. Per-Class Performance

The augmented ResNet50 classification report includes precision (fraction of predicted examples of a class that were correct), recall (fraction of actual examples of a class found), and F1-score (harmonic mean of precision and recall). The table below reproduces the exported class-level metrics.

Class	Precision	Recall	F1	Support
bumper_dent	0.88	0.74	0.80	19
bumper_scratch	0.67	0.50	0.57	12
door_dent	0.83	0.52	0.64	29
door_scratch	0.56	0.96	0.71	23
glass_shatter	0.91	1.00	0.95	21
head_lamp	0.85	0.85	0.85	20
tail_lamp	0.95	0.90	0.93	21

The macro averages were precision 0.81, recall 0.78, and F1-score 0.78. Glass shatter had perfect recall (21 of 21 examples), while bumper scratch had recall 0.50 (6 of 12) and door dent had recall 0.52 (15 of 29). The weaker categories warrant targeted data review rather than relying only on overall accuracy.

6.1 Class support and uncertainty

Per-class test support ranges from 12 to 29 images, so a few changed predictions can substantially alter class-level percentages. For example, one bumper-scratch example represents 8.3 percentage points of that class's recall. A larger external test set or repeated evaluations would better characterize performance stability.

7. Confusion Matrix and Failure Analysis

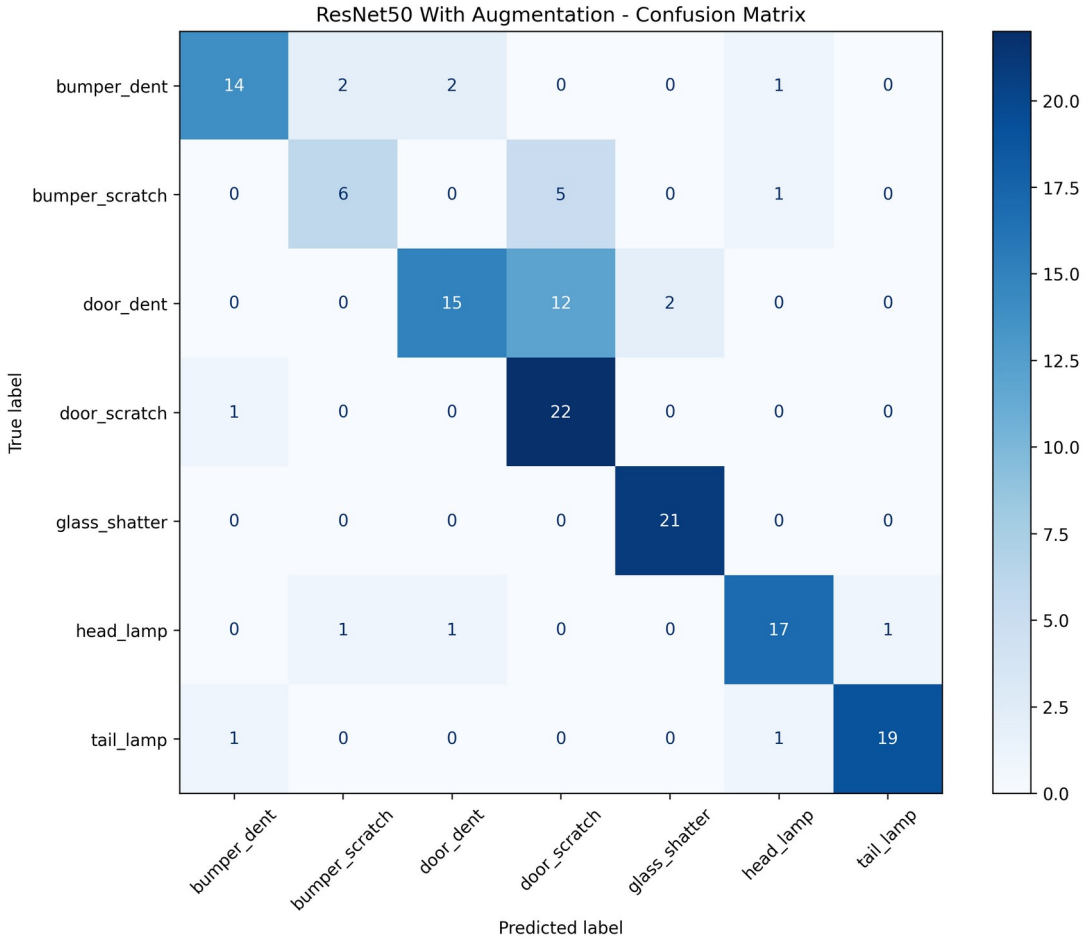


Figure 2. Confusion matrix for augmented ResNet50; rows indicate actual classes and columns predicted classes.

The largest visible confusion is door dent predicted as door scratch (12 images). Five bumper-scratch images were also predicted as door scratch. These patterns suggest that the classifier sometimes relies on similar local texture or reflection cues rather than reliably separating physical deformation from surface abrasion.

7.1 Visual examples of errors

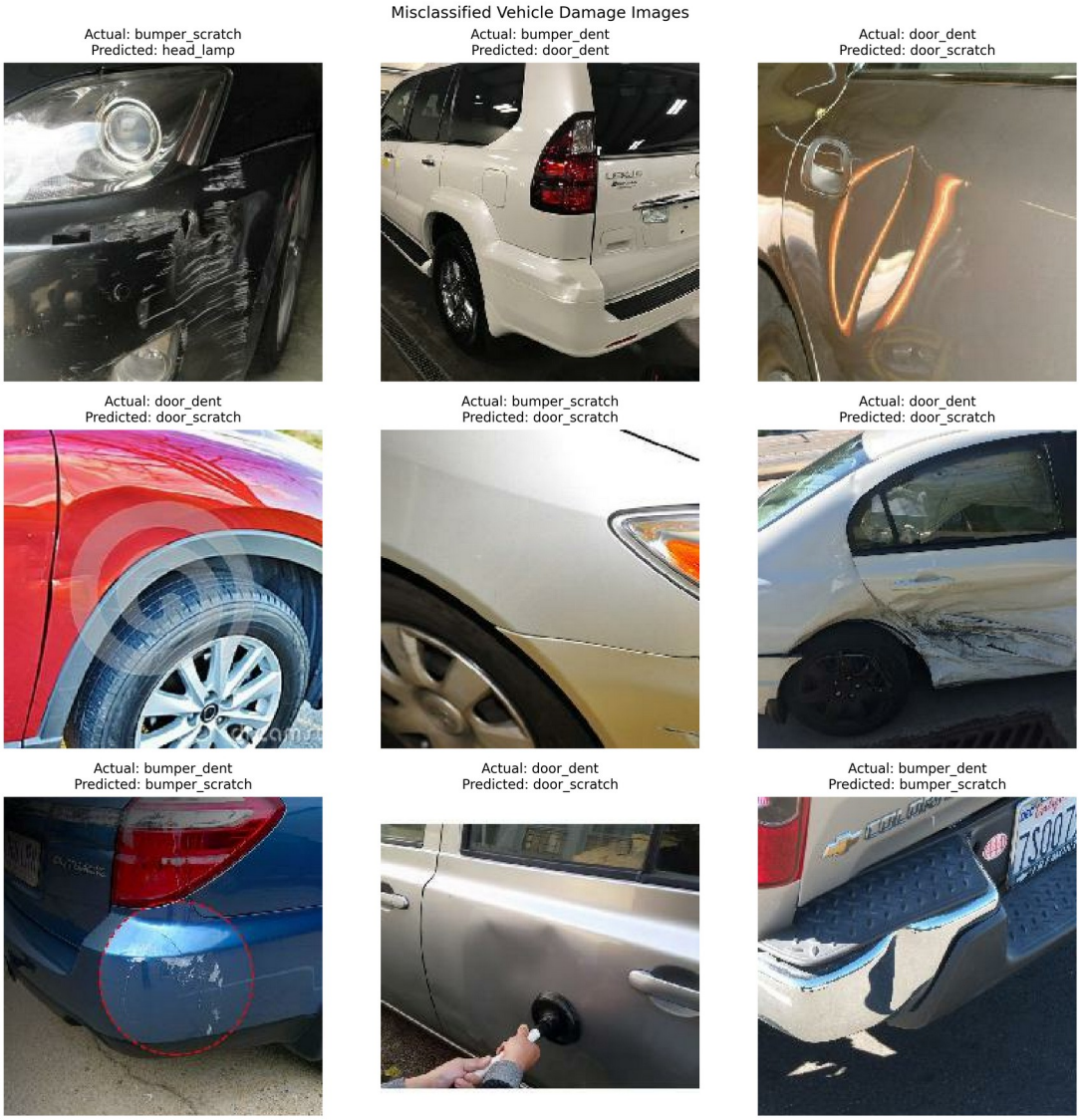


Figure 3. Examples of misclassified test images with true and predicted categories, as exported from the notebook. The augmented model made 31 errors among 145 test images.

The illustrated cases include bumper scratches predicted as head-lamp or door-scratch damage and door dents predicted as door scratches. Reflective surfaces, camera perspective, and weak or localized visual evidence are plausible contributors, but these explanations are qualitative hypotheses rather than verified causal findings.

7.2 Practical implications

For any real inspection workflow, these predictions should support—not replace—human judgment. A confidence threshold, rejection option for unfamiliar images, and an explicit quality-control step would be useful. Further testing should examine vehicles, cameras, and environmental conditions absent from the training data.

8. Limitations, Future Work, and Conclusion

8.1 Limitations

The labeled dataset is modest in size and contains duplicated references and unknown labels that require careful filtering. Category counts are unequal, and the test set is small. Only one recorded held-out split is available. The pretrained models were not established as a fully controlled augmentation ablation, and the original dataset provenance was not included with the uploads. No external validation, uncertainty calibration, or localization evaluation was conducted.

8.2 Future work

• Collect more diverse labeled examples, particularly door dents and bumper scratches.

• Standardize preprocessing, seed selection, training schedules, and evaluation for a controlled augmentation study.

• Fine-tune selected ResNet50 layers and compare with EfficientNet or MobileNet.

• Consider region-of-interest detection or segmentation for small, localized damage.

• Evaluate class-balanced metrics, calibration, and out-of-distribution robustness on an independent dataset.

8.3 Conclusion

Across the three recorded models, pretrained ResNet50 substantially outperformed the baseline CNN. The highest observed test accuracy was 78.62% for augmented ResNet50, with macro F1-score 0.78. Failure analysis highlights confusion among visually similar dent and scratch categories. The project provides a reproducible starting point for vehicle damage classification, while its limited sample size and non-identical experimental runs motivate more rigorous validation before deployment.

References and source notes

[1] K. He, X. Zhang, S. Ren, and J. Sun, “Deep Residual Learning for Image Recognition,” Proceedings of CVPR, 2016. (ResNet architecture.)

[2] TensorFlow / Keras documentation, “ResNet and ResNetV2” and “Image data augmentation layers.” (Software and preprocessing background.)

[3] Vehicle damage dataset: user-supplied archive.zip and data.csv. Original dataset author, publisher, license, and URL not supplied; verify and insert citation before submission.

[4] User-supplied experimental artifacts: final_model_comparison.csv, resnet50_aug_classification_report.csv, baseline_cnn_test_results.csv, training histories, and associated figures (October 2026).